NUS-ISS

Problem Solving Using Pattern Recognition

Deep learning: Before and After

Zhengqing Hu
Institute of System Science
National University of Singapore
Zq.hu@nus.edu.sg

Topic for today's lesson:

What are the problems we can solve pertaining to pattern recognition using deep learning?

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The Future Classrooms?



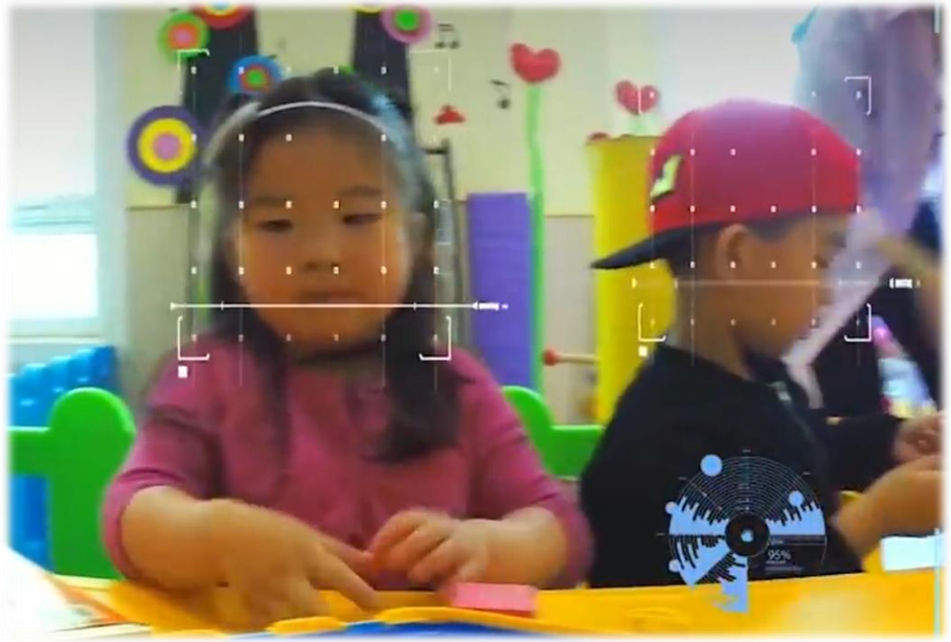
Source: https://www.youtube.com/watch?v=JMLsHI8aV0g

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The Future Classrooms?

Deep Learning Examples





- Translates brain signals into an attention score
- Recognize faces to take attendance
- Recognize gestures (e.g. raising hands, chatting, using mobile phones)
- Quiz students via robots
- Generate and inform health status with sensory data

Source: https://www.wsj.com/articles/chinas-efforts-to-lead-the-way-in-ai-start-in-its-classrooms-11571958181

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The Future Classrooms?

Satisfactory results



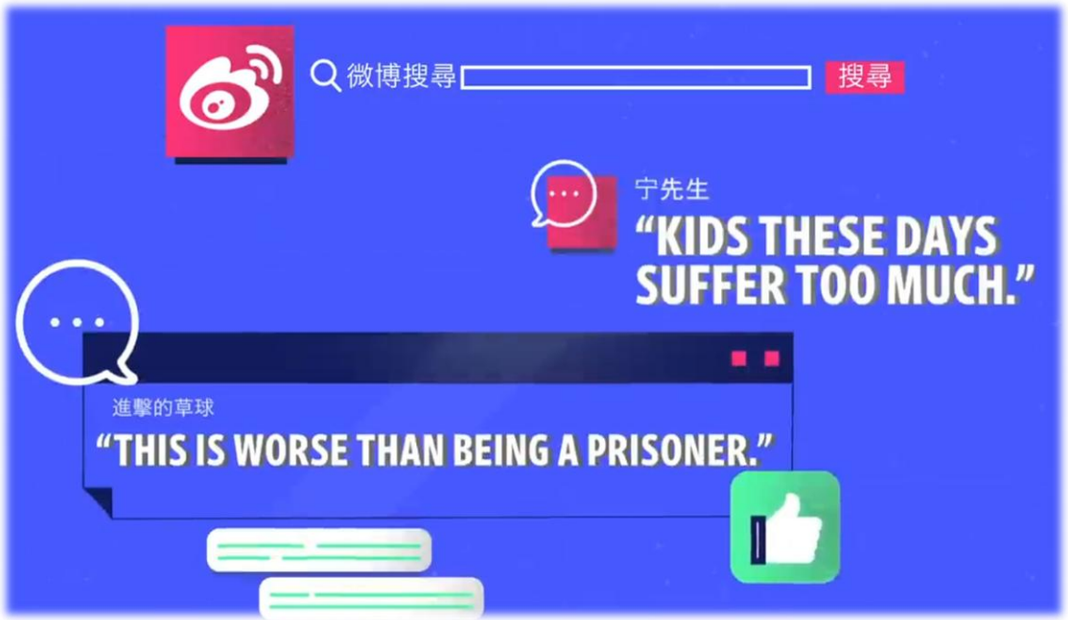
- Increased attention lead to improved grades
- Less work for teachers (automated attendance taking, less focus on observing and more focus on teaching)
- Monitoring by worried parents (some responsibility is shifted to the parents to discipline/ educate their own children instead of the teacher bearing full responsibility)

Source: https://www.wsj.com/articles/chinas-efforts-to-lead-the-way-in-ai-start-in-its-classrooms-11571958181

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The Future Classrooms?

Issues of such applications





Source: https://www.wsj.com/articles/chinas-efforts-to-lead-the-way-in-ai-start-in-its-classrooms-11571958181

- No data privacy
- Chances of not being to obtain parent consent
- Technologies might not be accepted by society as a whole (undesirable social media feeds)
- Concerns that students will feel too pressured under surveillance

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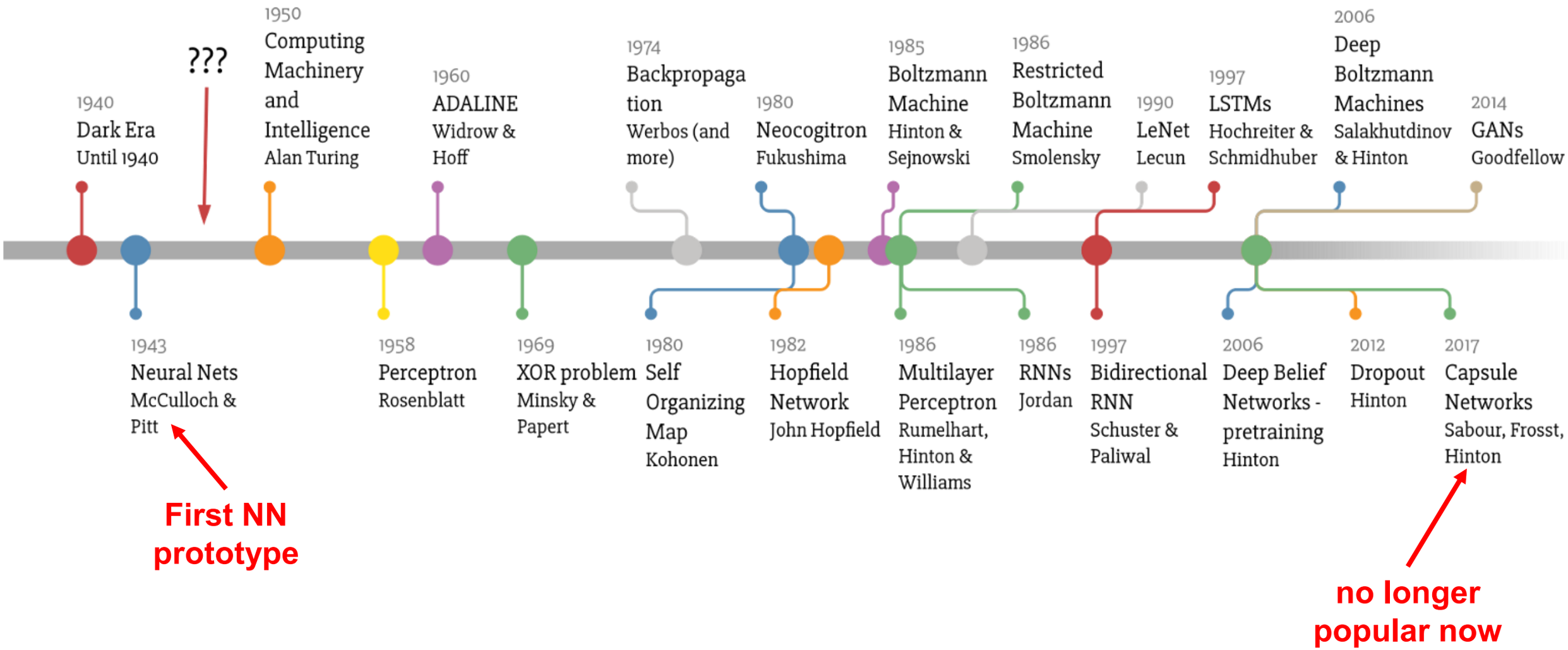
Deep learning: The Before

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Time line

Of deep learning

Many methods will sink throughout until there is a convergence (i.e. standards produced)



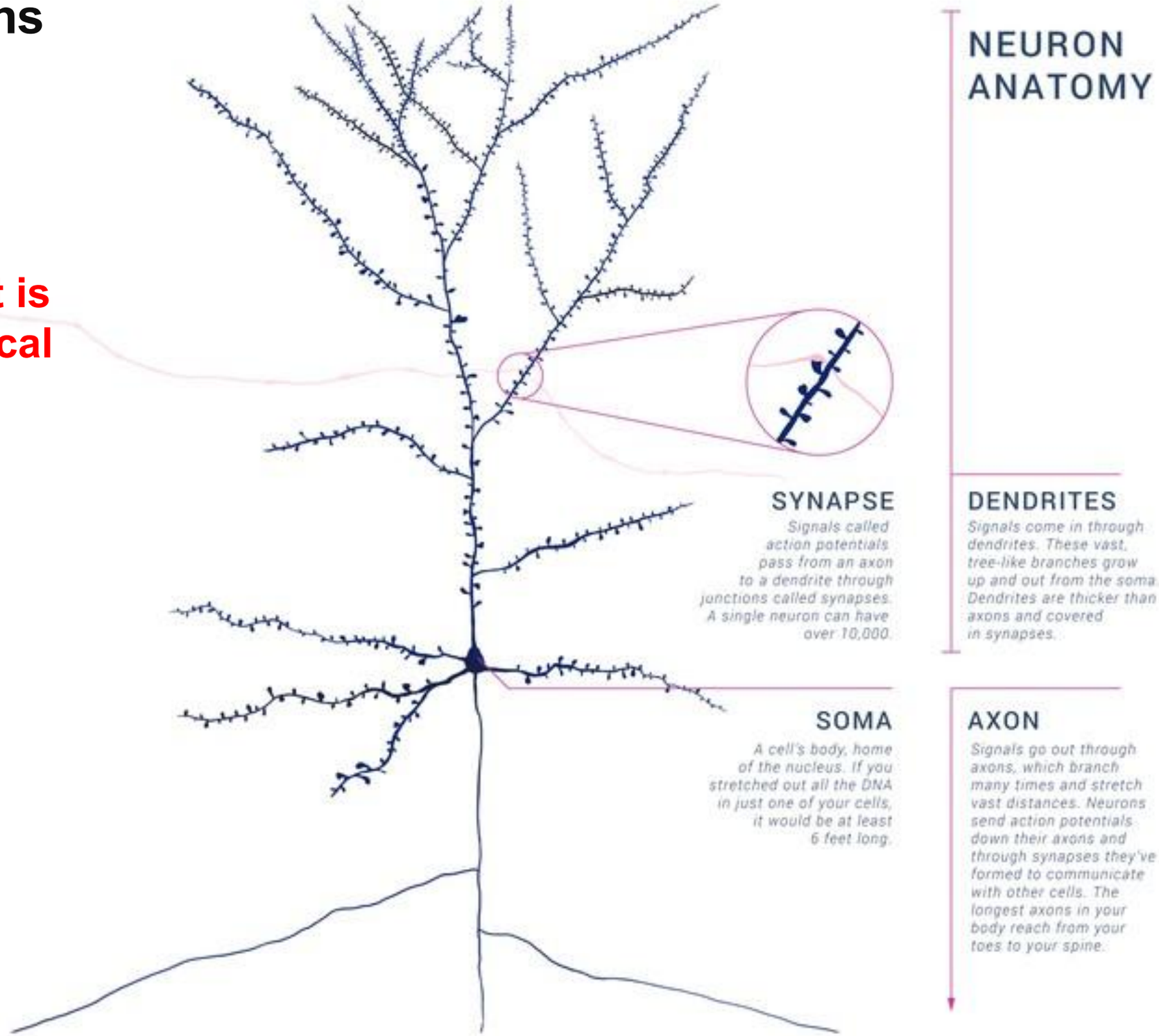
Source: https://medium.com/@faviovazquez

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Biological neurons

Simplified illustration

Neural Net concept is inspired by biological neurons!

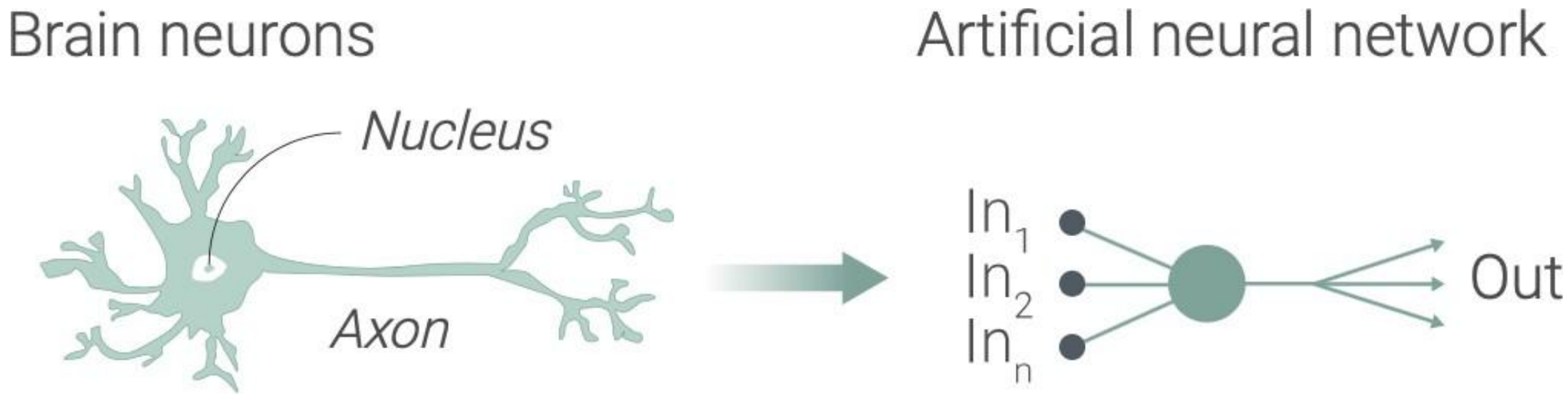


Source: https://en.wikipedia.org/wiki/File:Anatomy_of_a_Neuron_with_Synapse.png

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The evolution

From neurons to neural network



For ANN to be powerful, we will need more neurons, just like our brain!

Source: https://multimedia.scmp.com/news/china/article/2166148/china-2025-artificial-intelligence/index.html?src=follow-chapter

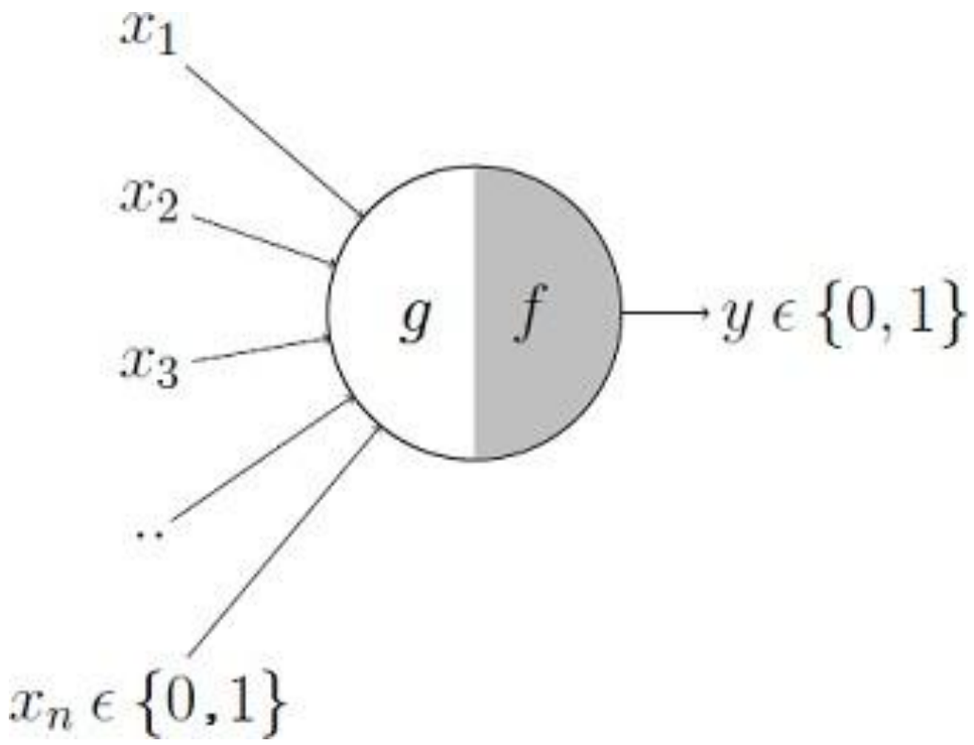
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The first artificial neuron

McCulloch-Pitts Neuron

- By Warren McCulloch (neuroscientist) and Walter Pitts (logician) in 1943

Note that the inputs (i.e. x_n) in here are only 0 or 1 (true or false)



$$g(x_1, x_2, x_3, ..., x_n) = g(\mathbf{x}) = \sum_{i=1}^{n} x_i$$

$$y = f(g(\mathbf{x})) = 1 \quad if \quad g(\mathbf{x}) \geq \theta$$
$$= 0 \quad if \quad g(\mathbf{x}) < \theta$$

g(x) = sum of all inputs
f(g(x)) = output defined by the function

Source: https://en.wikipedia.org/wiki/File:Anatomy_of_a_Neuron_with_Synapse.png

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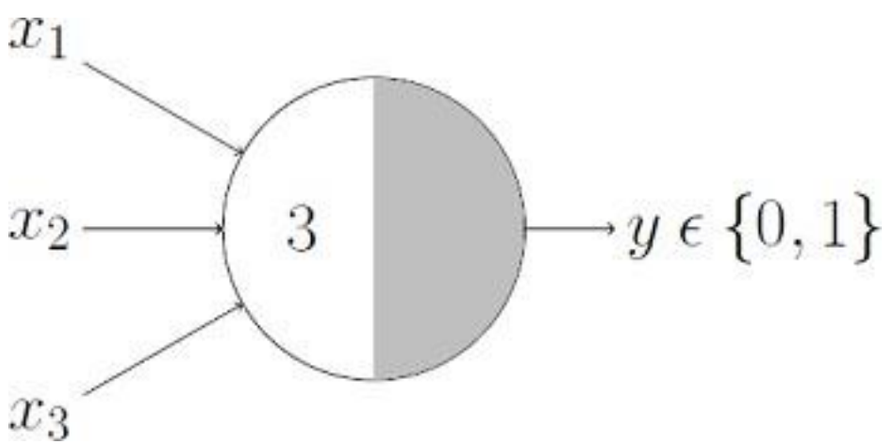
The first artificial neuron

McCulloch-Pitts Neuron

- Can be used to represent a few Boolean functions

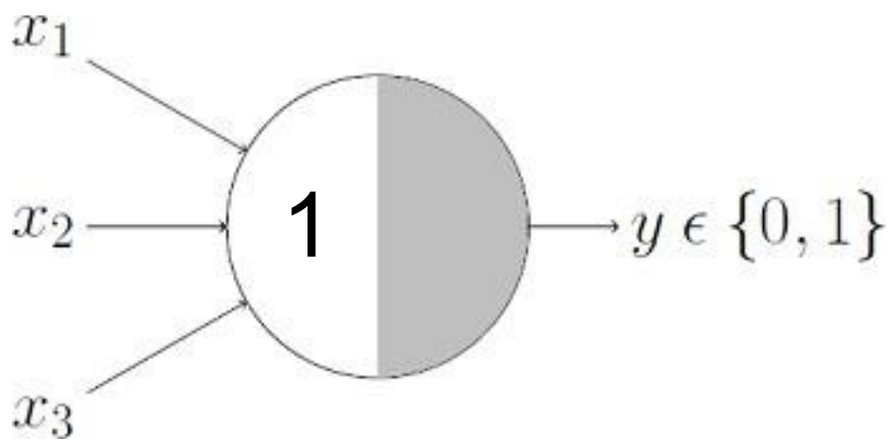
Interesting but NOT useful in real life application!

AND function



$y = 1$ if $x_1 + x_2 + x_3 \geq 3$

OR function



$y = 1$ if $x_1 + x_2 + x_3 \geq 1$

Source:https://towardsdatascience.com/mcculloch-pitts-model-5fdf65ac5dd1

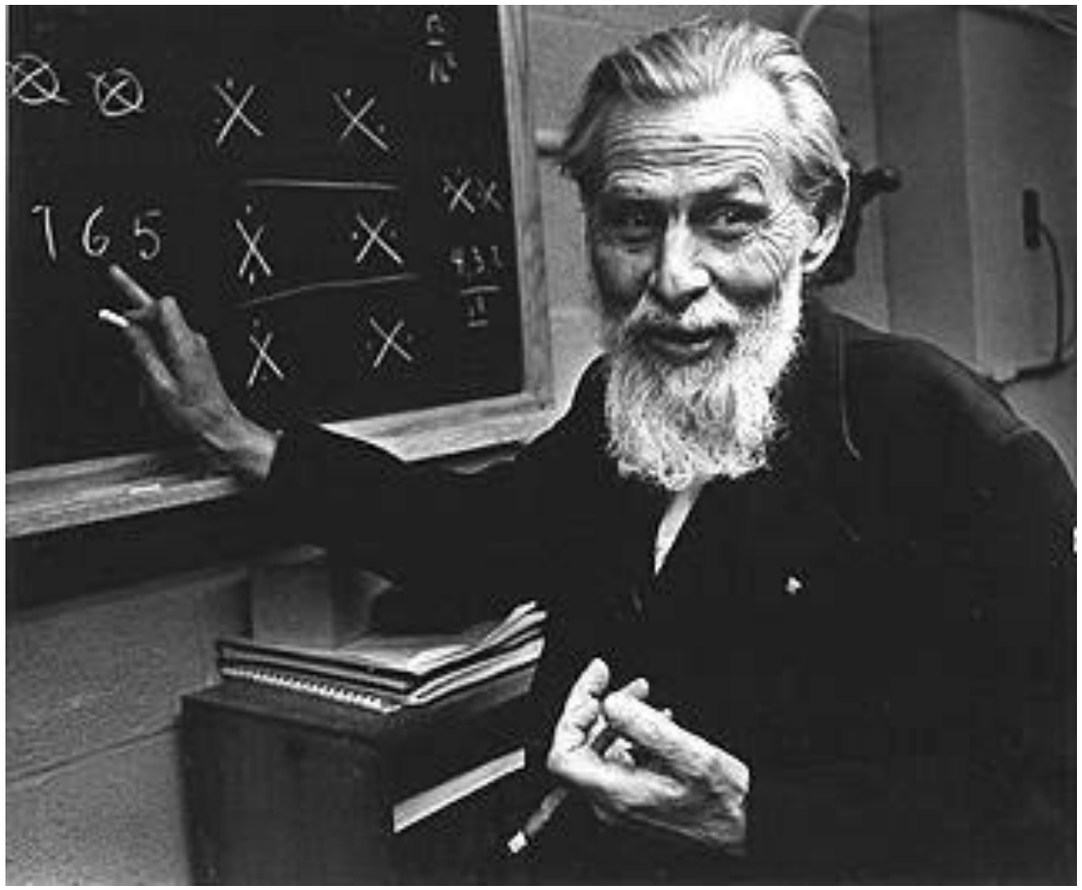
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The first artificial neuron

McCulloch-Pitts Neuron

- Inputs accepts only boolean values
- No learning algorithm to update the neurons

No learning = NOT true AI !!!





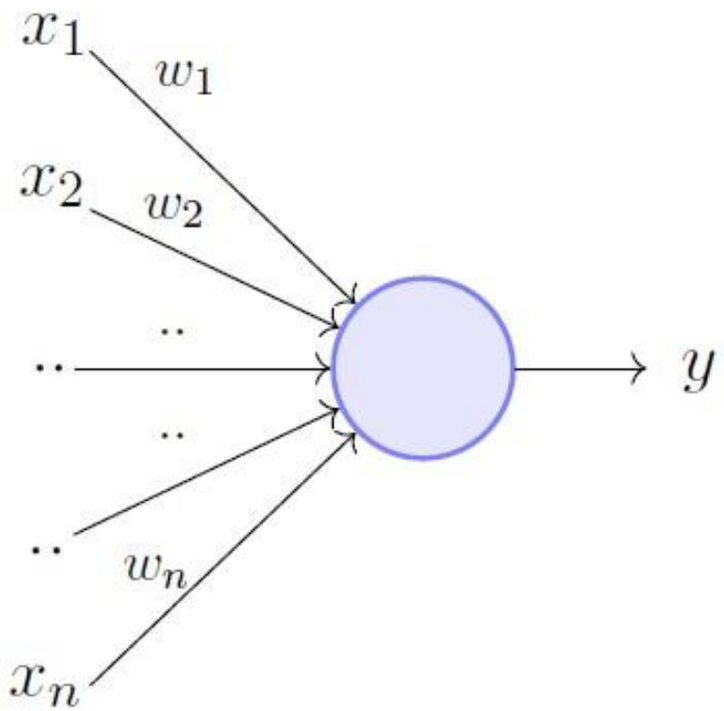
Left: Warren S. McCulloch. Right: Walter H. Pitts Jr.

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The improved artificial neuron

Perceptron

w_n are the weights,
Θ is the threshold value



- By Frank Rosenblatt, refined by Minsky and Papert
- Support real inputs, not just boolean values

$$y = 1 \quad if \sum_{i=1}^{n} w_i * x_i \geq \theta$$

$$= 0 \quad if \sum_{i=1}^{n} w_i * x_i < \theta$$

Rewriting the above,

$$y = 1 \quad if \sum_{i=1}^{n} w_i * x_i - \theta \geq 0$$

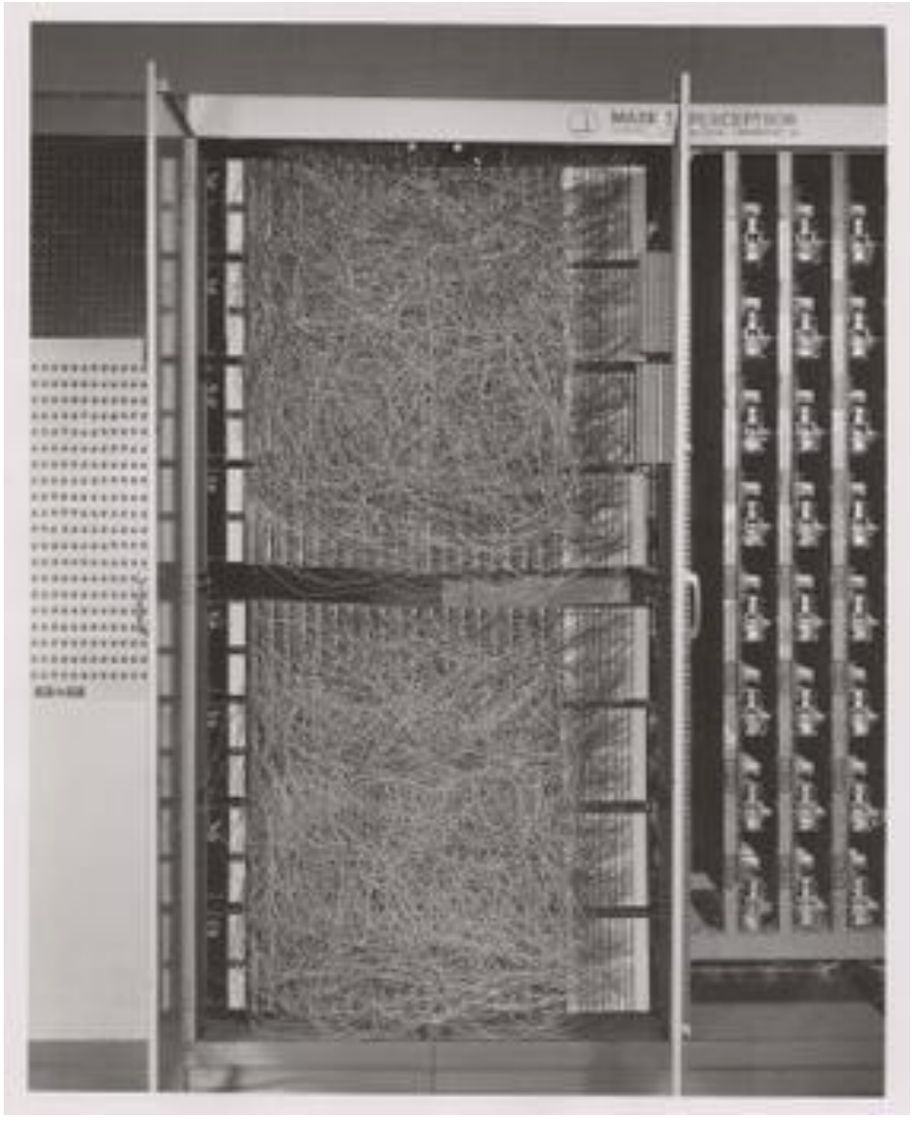
$$= 0 \quad if \sum_{i=1}^{n} w_i * x_i - \theta < 0$$

Source: https://towardsdatascience.com/perceptron-the-artificial-neuron-4d8c70d5cc8d

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The improved artificial neuron

Perceptron



Source: 'Mark I Perceptron at the Cornell Aeronautical Laboratory', hardware implementation of the first Perceptron (Source: Wikipedia / Cornell Library)

- Rosenblatt's achievement: artificial neurons could actually learn from data
- He came up a supervised learning algorithm!
- He implemented Perceptron in custom hardware, which can learn to classify simple shapes correctly with 20x20 pixel-like inputs

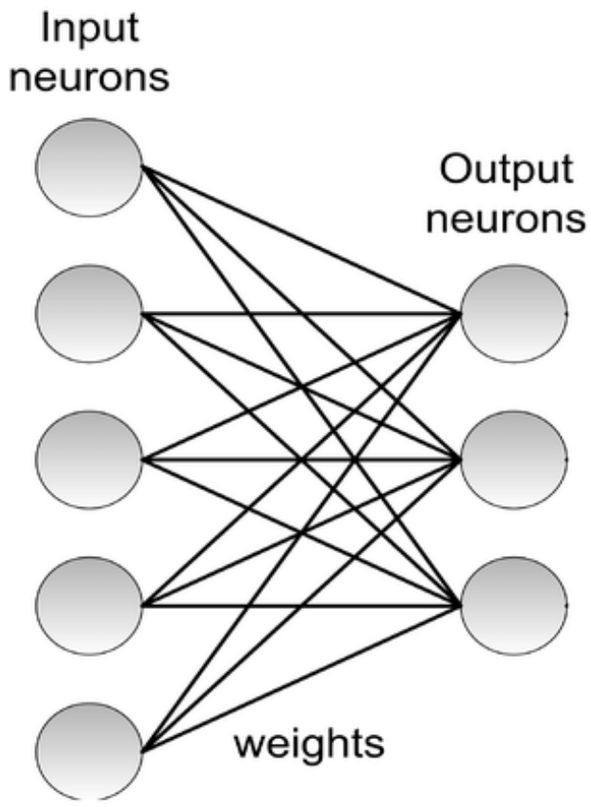
The perceptron method can learn from past data!

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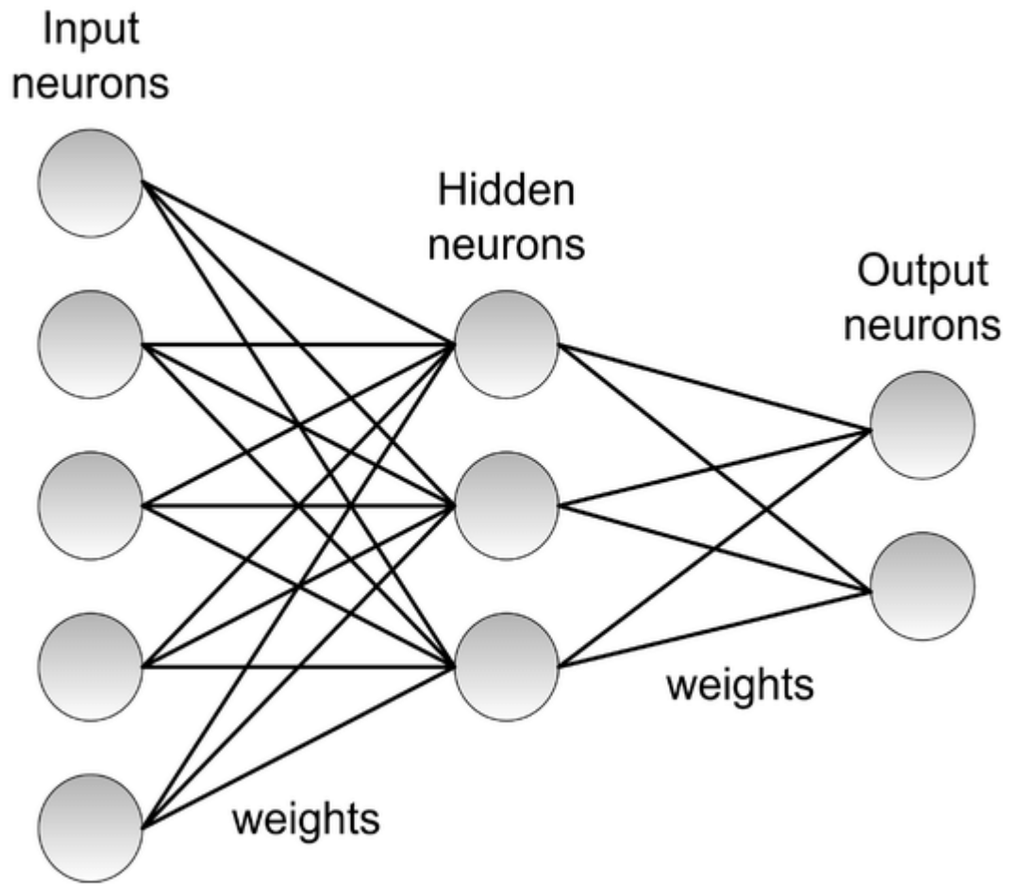
The improved artificial neuron

Perceptron

Note that the neurons are aligned into a single line for each layer (can be represented with a 1D array)



Single layer perceptron



Multi-layer perceptron

Source: 'Mark I Perceptron at the Cornell Aeronautical Laboratory', hardware implementation of the first Perceptron (Source: Wikipedia / Cornell Library)

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The first AI winter

XOR affair



Source: https://amethix.com/2018/06/ai-winter-is-coming/

- Marvin Minsky, founder of MIT AI lab, and Seymour Papert, director of the lab in 1969 published a book named 'Perceptrons'
- They showed that a single perceptron cannot do XOR
- Multiple layers of Perceptron (aka MLP) can do XOR, but the proposed learning algorithm does not work for that!
- Here comes the winter ...

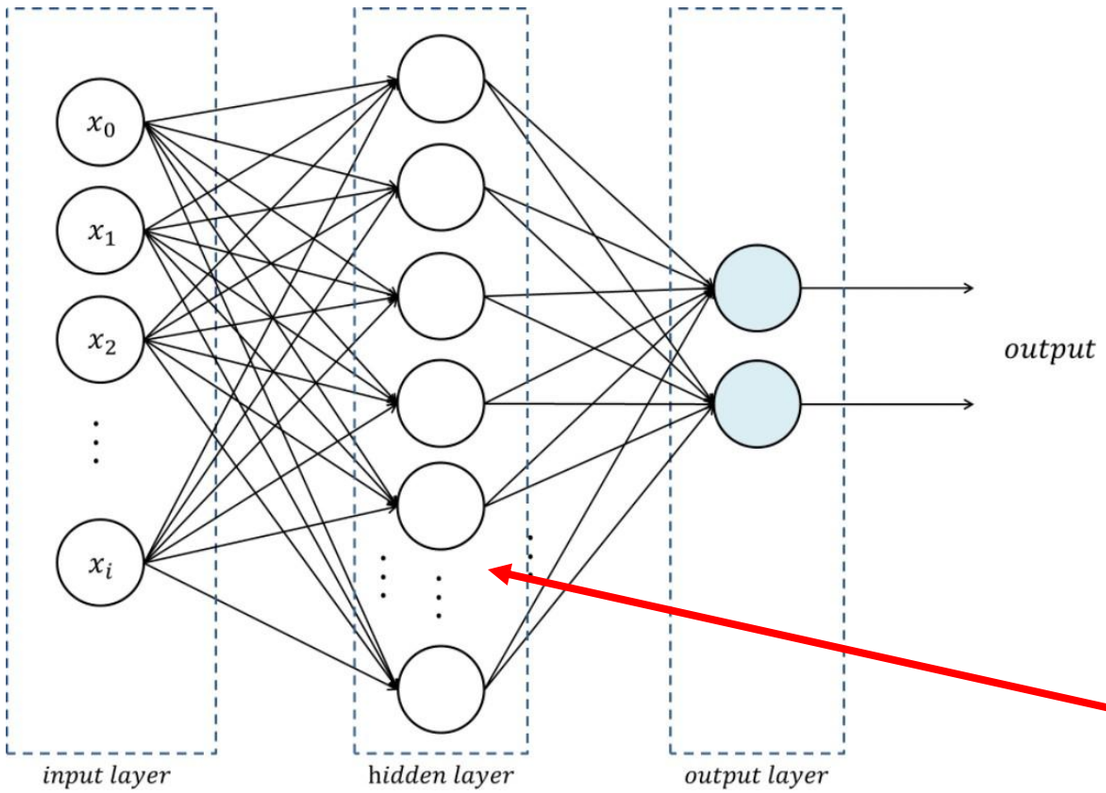
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Where is the learning algorithm?

The thaw of AI winter

Backpropagation = a learning algorithm that makes the Neural Network (NN) learn!

How? It minimizes the error in the output of a neural network by adjusting the weights of the network



Source: https://www.cc.gatech.edu/~san37/post/dlhc-fnn/

- Multilayer layers of perceptron should work, but need learning algorithm
- Between 60s and 80s, several researchers separately derived the solution, but few people knew
- In 1986, Rumelhart, Hinton Williams published a method in Nature
- They called the learning procedure "backpropagation"

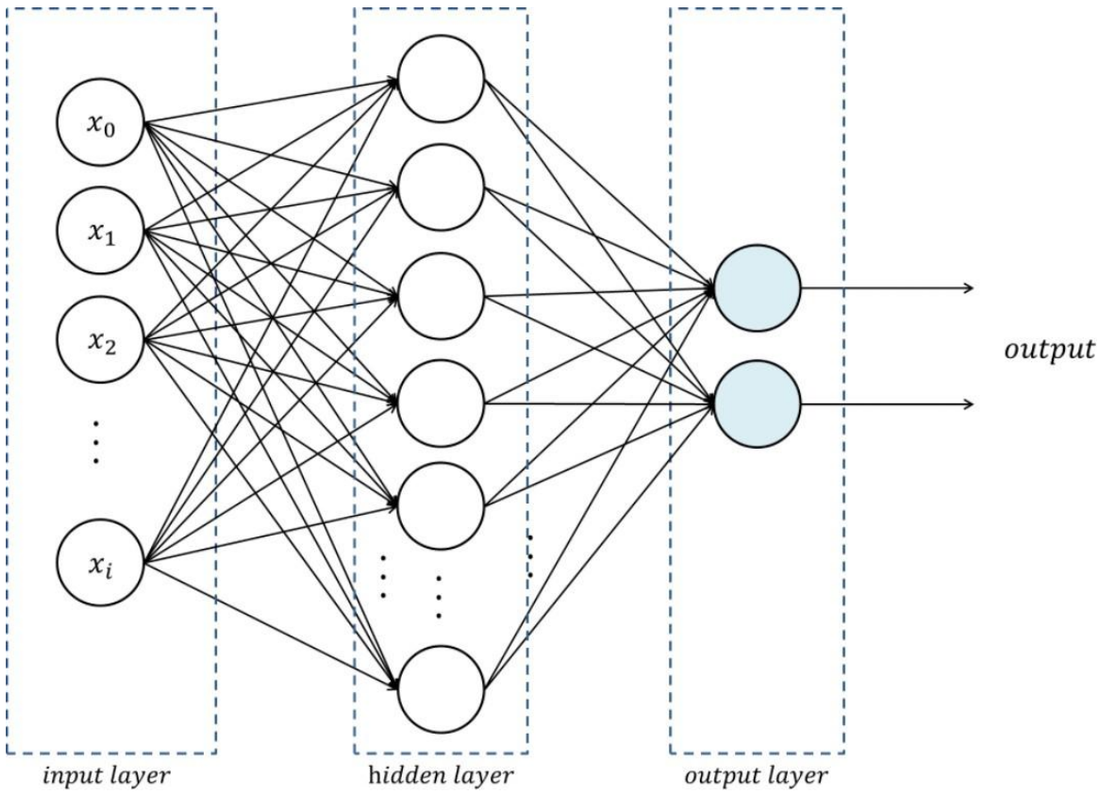
Can have more than 1 hidden layer; more layers = deep NN

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Where is the learning algorithm?

The thaw of AI winter

Question:
How can we fit an image (square matrix) into a NN?



Source: https://www.cc.gatech.edu/~san37/post/dlhc-fnn/

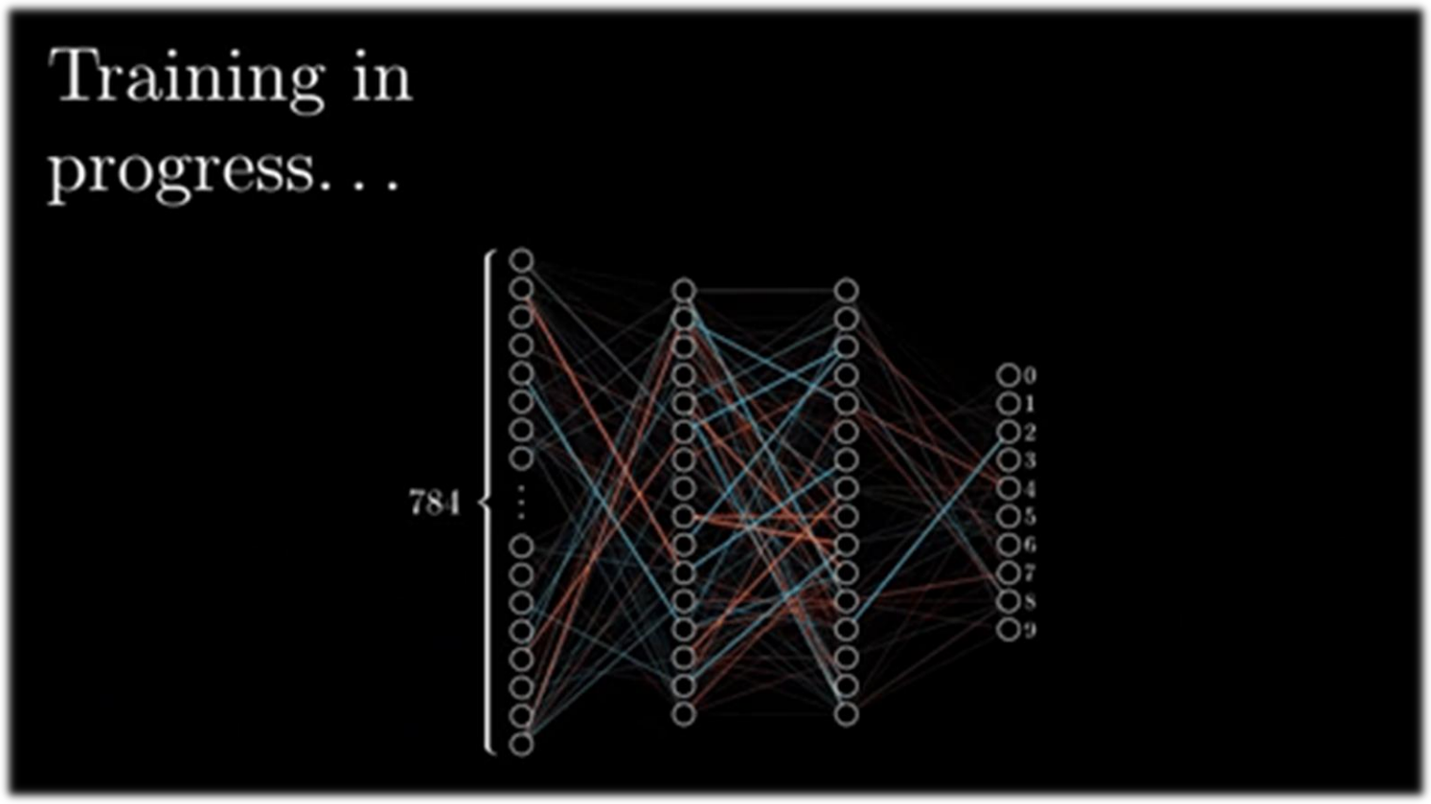
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Where is the learning algorithm?

The thaw of AI winter

Visual Illustration of Backpropagation Process:

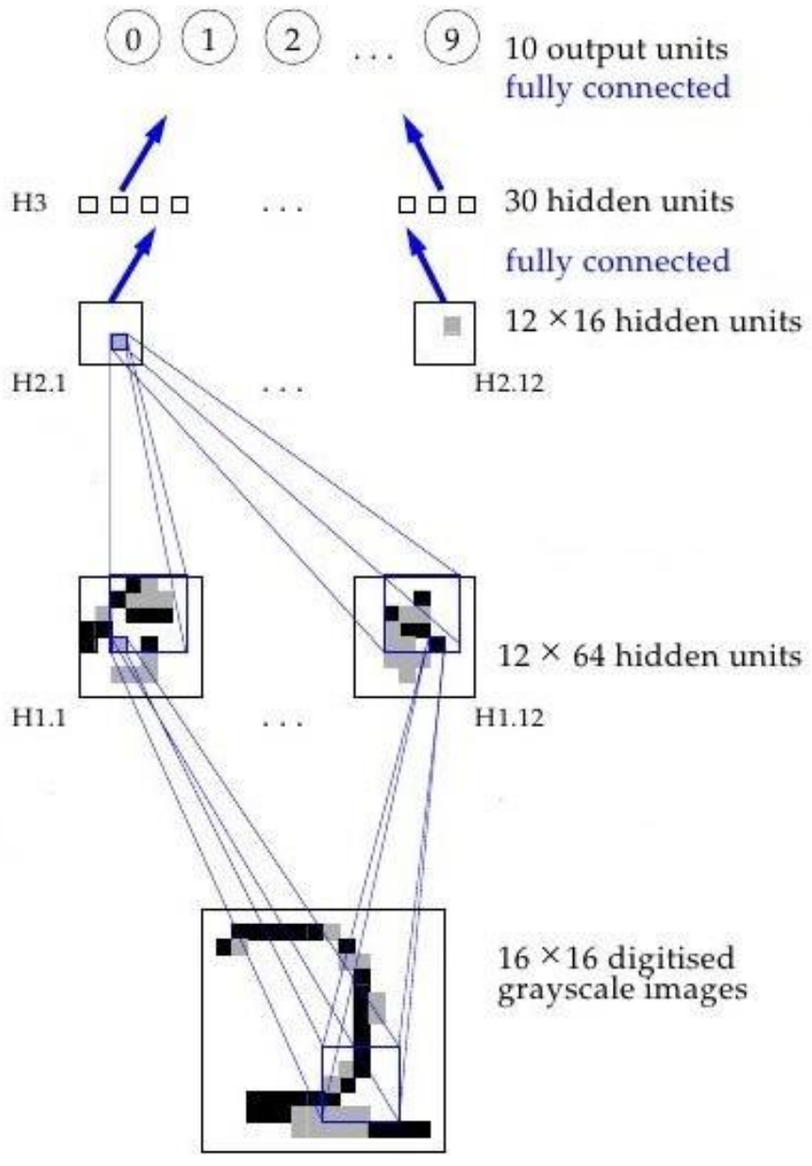


Forward Pass → Compute Error → Backward Pass → Update Weights

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Year 1989

Annus mirabillis, sort of



Source: http://www.andreykurenkov.com/writing/ai/a-brief-history-of-neural-nets-and-deep-learning-part-2/

- Multilayer feedforward networks are proved to be universal approximators (i.e. able to approximate any functions)
- LeNet was proposed and put into actual significant use: recognizing numbers
- It is a neural net + convolutional layers (weight sharing)

Convolutions = performing filtering

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Another winter dawns

by backpropagation



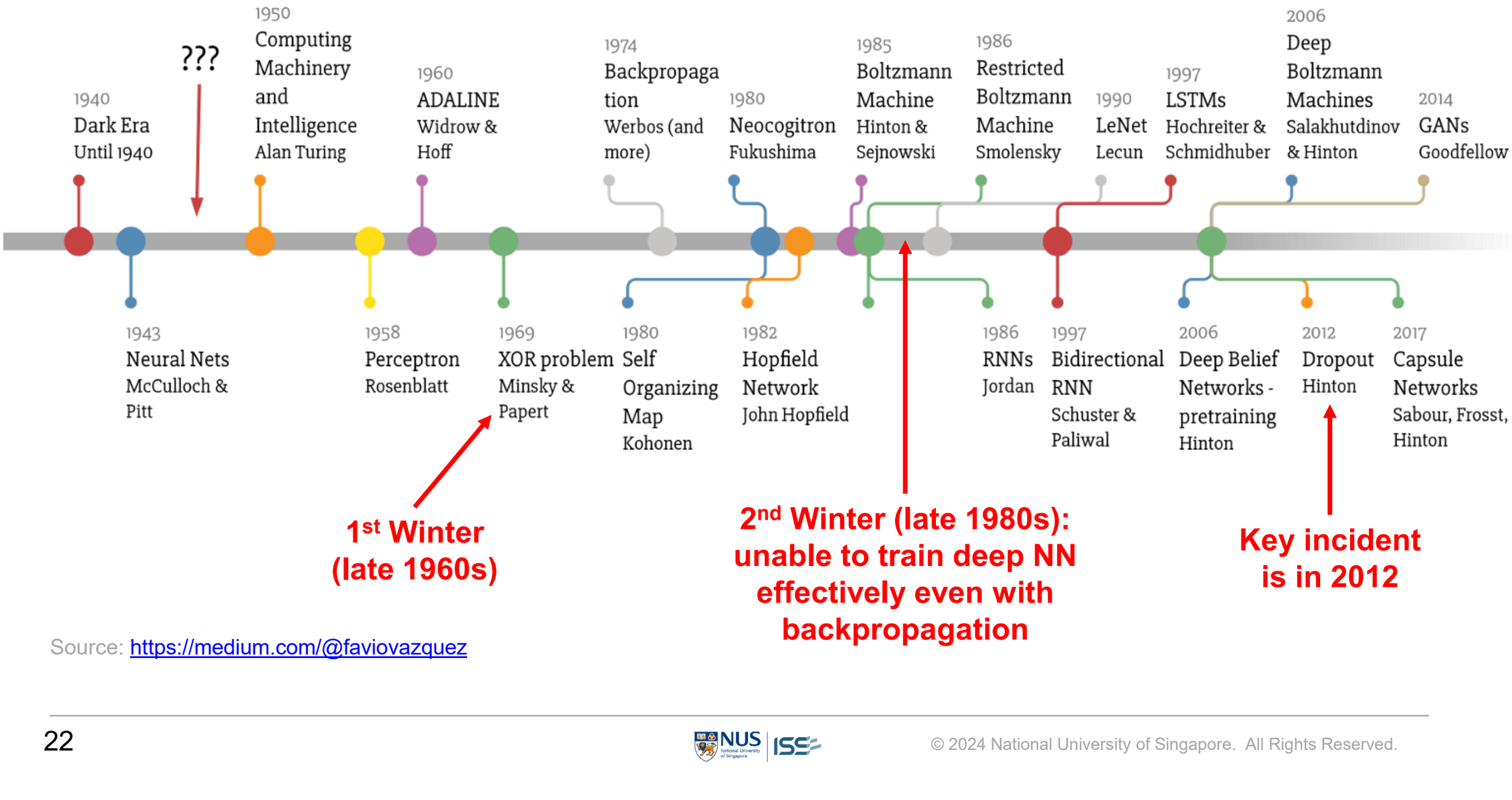
Source: https://amethix.com/2018/06/ai-winter-is-coming/

- By late 80s, already knew that deep neural net was hard to train (i.e. net unable to coverage / no improvement in the classification results / the net cannot conclude anything)
- Deep neural nets trained with backpropagation did not work very well, not as well as nets with fewer layers
- Support vector machine (SVM) came into fashion
- Random forests, with lovely mathematical theory behind proved to be effective

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Time line

Of deep learning



Source: https://medium.com/@faviovazquez

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ImageNet

by Stanford Vision Lab



1.2 million images, 1000 object categories

Source: https://gluon-cv.mxnet.io/build/examples_datasets/imagenet.html

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Competition
on ImageNet

1.2 million images:
- **1 million for training**
- **150k for testing**
- **50k for validation (final testing)**

- ILSVRC: ImageNet Large Scale Visual Recognition Challenge
- Started from 2010; teams evaluate their algorithms on given data set, compete to achieve highest accuracy on visual recognition tasks.
- ILSVRC training dataset: 1000 object categories, 1.2 million images



Source: http://www.image-net.org

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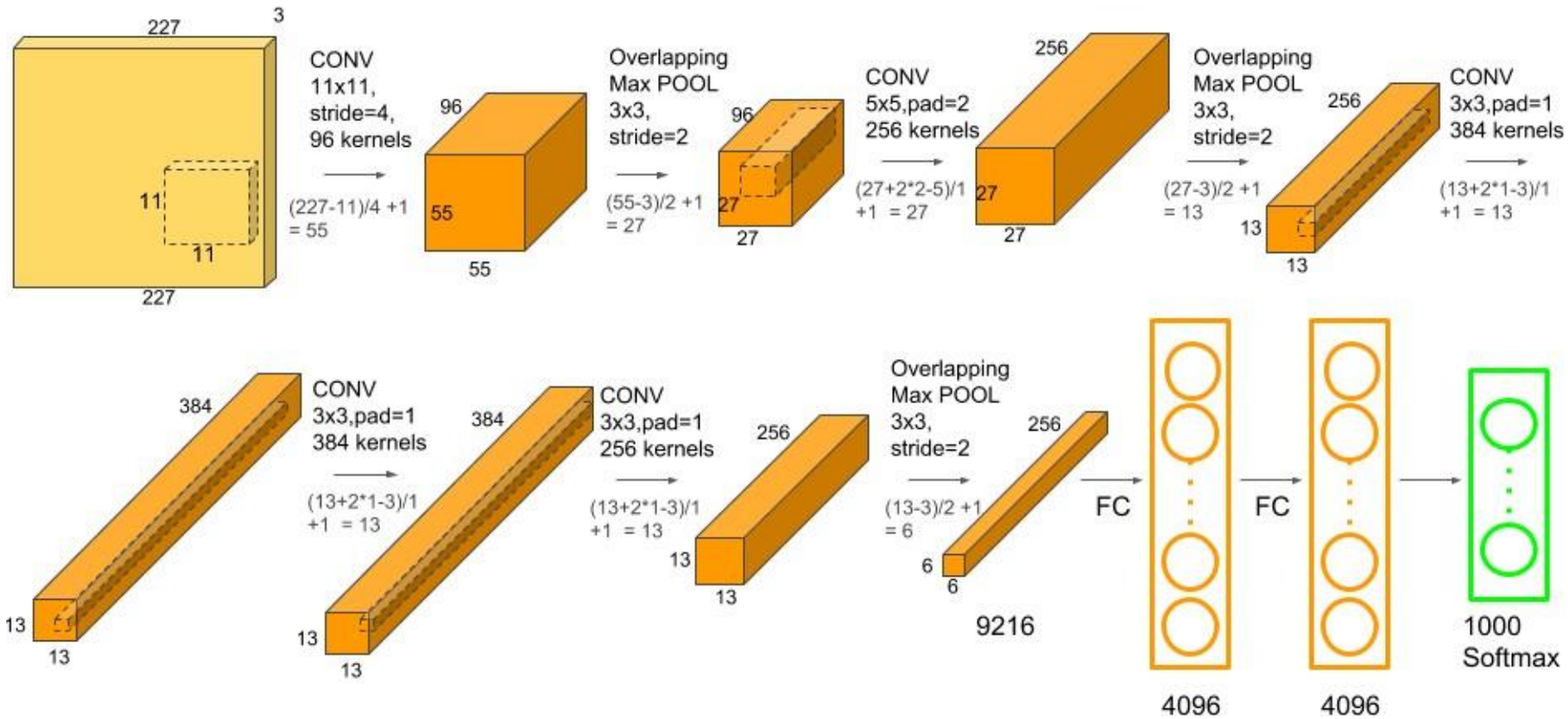
ILSVRC2012

AlexNet

- Team Hinton entered the competition

AlexNet is a Convolutional Neural Network (CNN) model; similar to LeNet

- They achieved an error rate of 15.3%, far far better than the next closest: 26.2%



Where are the input, CNN, hidden and output layers?

Source: https://neurohive.io/en/popular-networks/alexnet-imagenet-classification-with-deep-convolutional-neural-networks/

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AlexNet

Key of success?

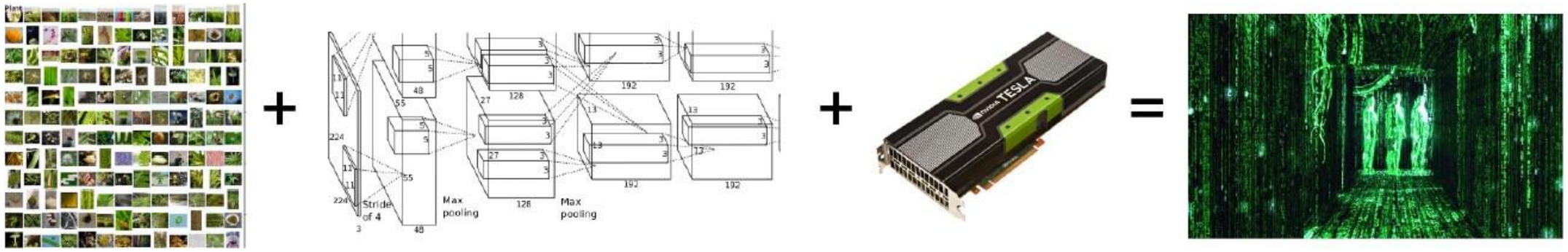
Activation function decides, whether a neuron should be activated or not, by calculating the weighted sum and further adding bias with it

The main purpose of the activation function is to introduce non-linearity into the output of a neuron

- The net structure is not the most important point
- Use of Rectified Linear Unit (ReLU) activation function
- Use of dropout
- GPU implementation (through CUDA); 20x time difference!

Why need non-linear activation function ???

Deep learning computer vision recipe



Lots of data + Deep convolutional neural network + Backprop on GPU = Vision

Source: http://www.computervisionblog.com/2015/05/deep-learning-vs-big-data-who-owns-what.html

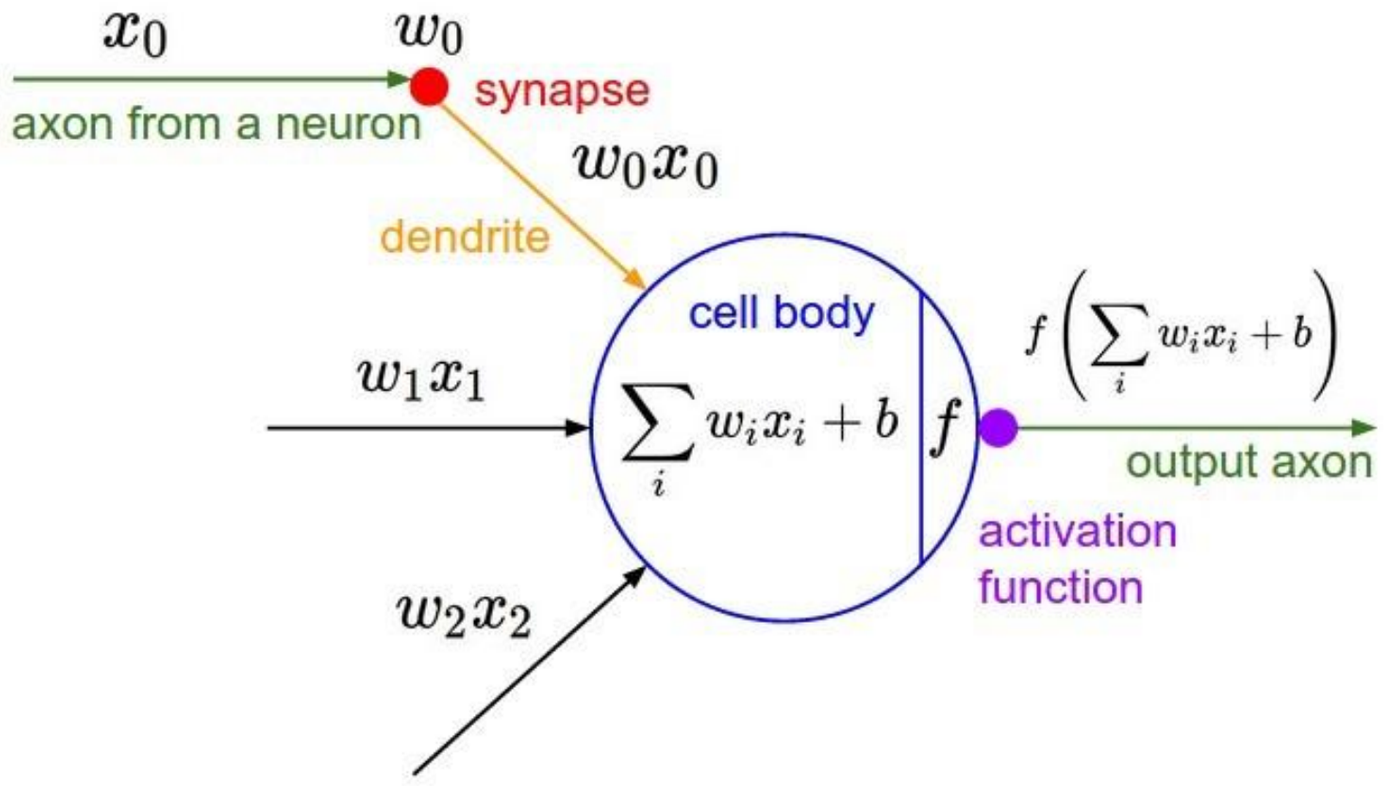
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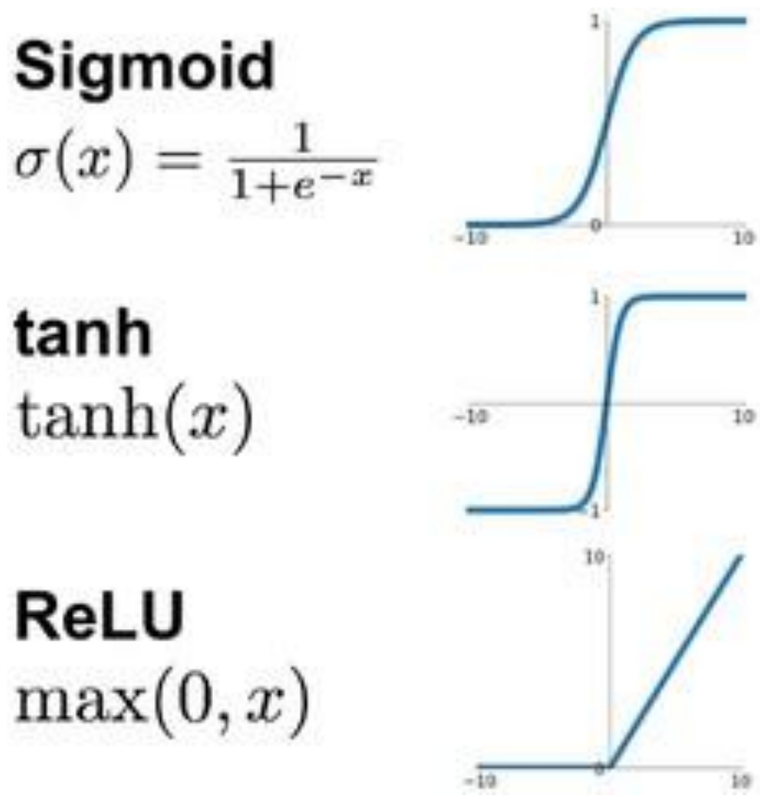
ReLU, as compared to others

Simple is better

ReLU is the fastest, simplest and most efficient as compared to the other 2 methods; now the default for activation function

- The zero value output from ReLU introduces sparsity representation (more zeros in each layer, only the important neurons contribute)
- ReLU is easier to calculate





Source: http://www.andreykurenkov.com/writing/ai/a-brief-history-of-neural-nets-and-deep-learning/

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ReLU, as compared to others

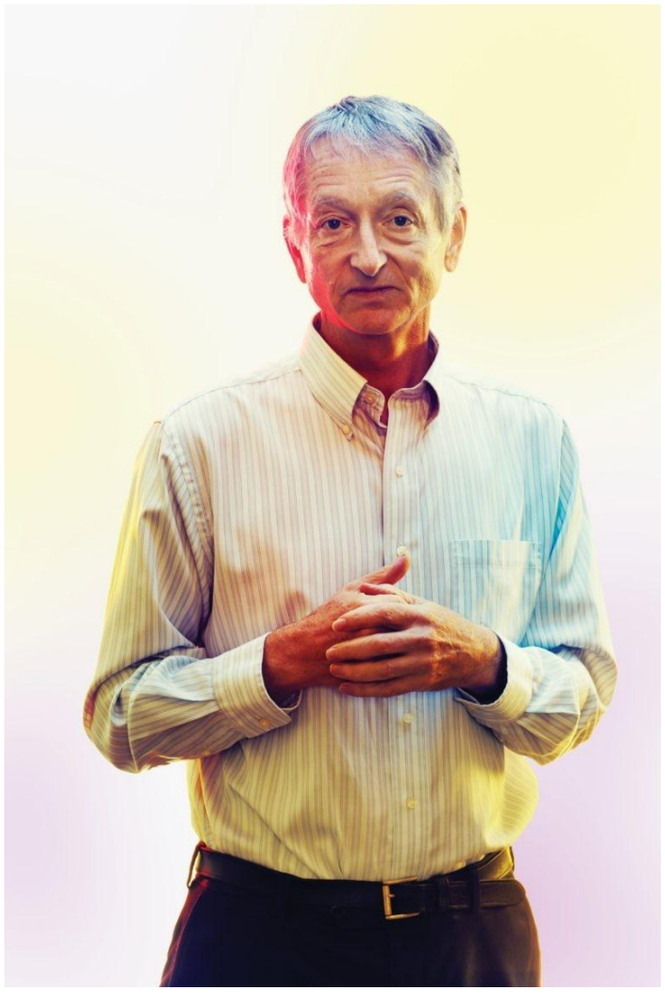
Simple is better

Property	ReLU	Sigmoid	Tanh
Output Range	(0, ∞)	(0, 1)	(-1, 1)
Computational Cost	Low (linear)	High (exponential)	High (exponential)
Vanishing Gradient Problem	No	Yes	Yes
Convergence	Fast convergence	Slower convergence	Faster than Sigmoid
Saturation Problem	No	Yes	Yes
Common Use Cases	Hidden layers in convolutional neural networks, feedforward neural networks	Output layer of binary classification problems	Hidden layers in feedforward neural networks, recurrent neural networks

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The big question

Why did backpropagation fail in the past?



Source: https://torontolife.com/tech/ai-superstars-google-facebook-apple-studied-guy/

- Labelled datasets were thousands times too small; deep learning applications generally require a lot of datasets!

- Computing power was millions times slower back then; computing is now much faster as compared to the past

- Initialized the weight in stupid ways; weights are all set to 0 or 1 in the past

- Used the wrong type of non-linearity for activation function

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Since then ...

High-level Neural Networks Library

theano

Keras

Caffe

DEEPLEARNING4J

Open-Source Machine Learning Libraries

TensorFlow

PYTORCH

mxnet

The Microsoft Cognitive Toolkit

Chainer

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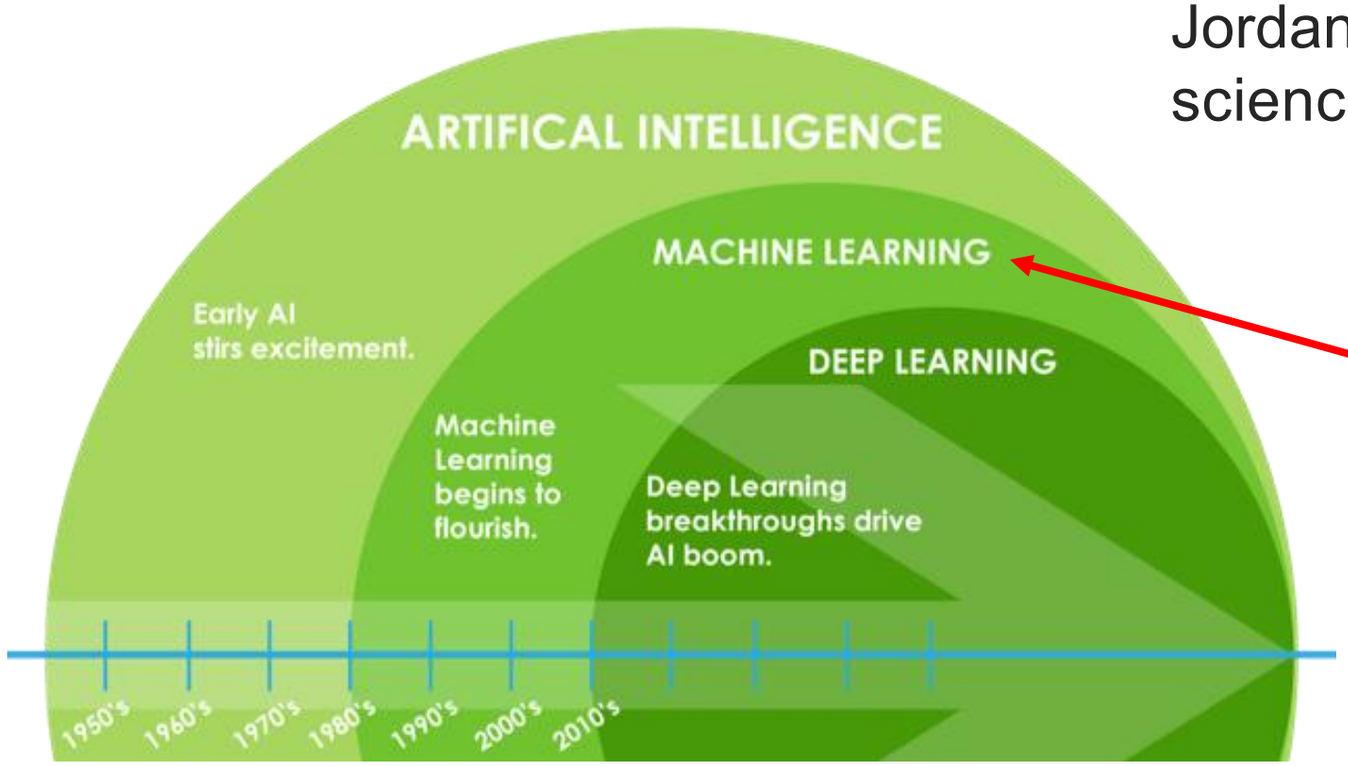
Deep learning: The After

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Confusion?

The differences among the few terms

- Arthur Samuel coined the term 'machine learning' in 1959 while at IBM
- Rina Dechter introduced the term deep learning in 1986
- Machine learning and statistics are closely related, thus Michael I. Jordan suggested the term 'data science' to refer to the overall field



Covered in past few modules!

Source: https://buzzrobot.com/difference-between-artificial-intelligence-machine-learning-and-deep-learning-ccfd779eca7b

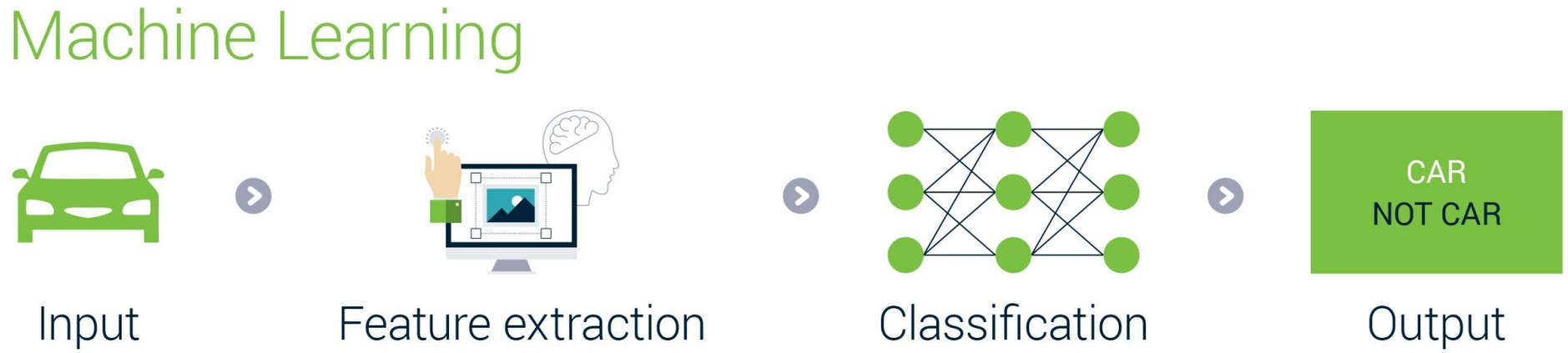
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Confusion?

The differences among the few terms

- Feature: a number or a vector that describes something about the input

What is the main difference between machine learning and deep learning ???



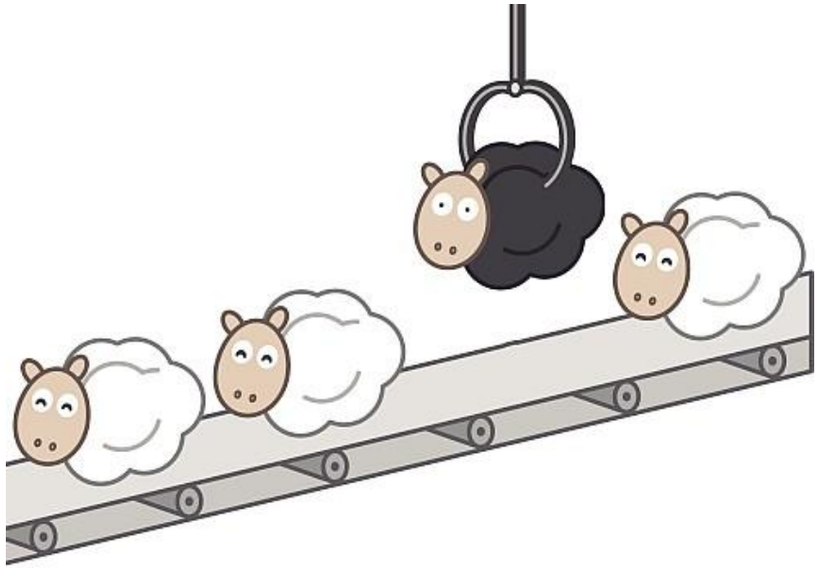
Source: https://verhaert.com/difference-machine-learning-deep-learning/

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Application

Three main categories

Identify



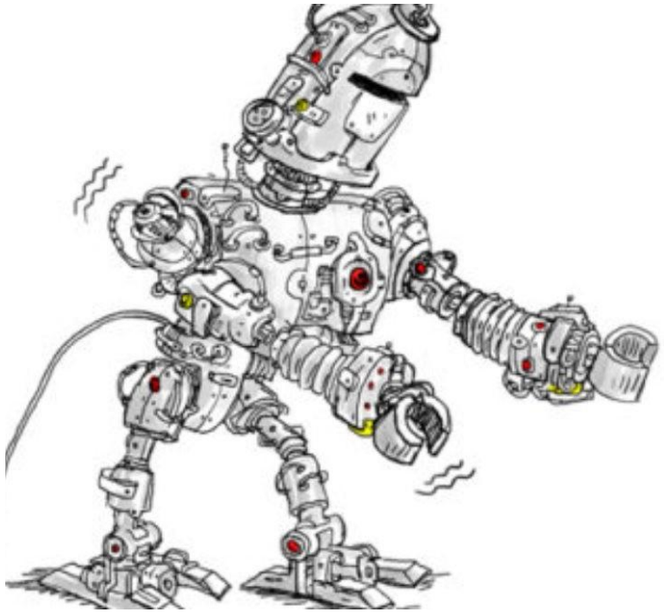
Source: http://blog.ss8.com

Create



Source: http://www.williammalone.com/articles/create-html5-canvas-javascript-drawing-app/

Act



Source: https://bitsandatoms.co/tag/reinforcement-learning/

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